

Platform Catalog Document

Project Title:

Mini Internal Developer Platform (Mini IDP)

Submitted By:

Sachin Bisen

Internship Organization:

Capgemini

Project Domain:

T1 Cloud

1. Introduction

The Mini Internal Developer Platform (Mini IDP) is a cloud-native DevOps automation platform developed as part of the Capgemini Sprint Project under the Cloud Domain.

The platform simplifies application deployment, infrastructure provisioning, monitoring, and CI/CD automation for developers using modern DevOps practices.

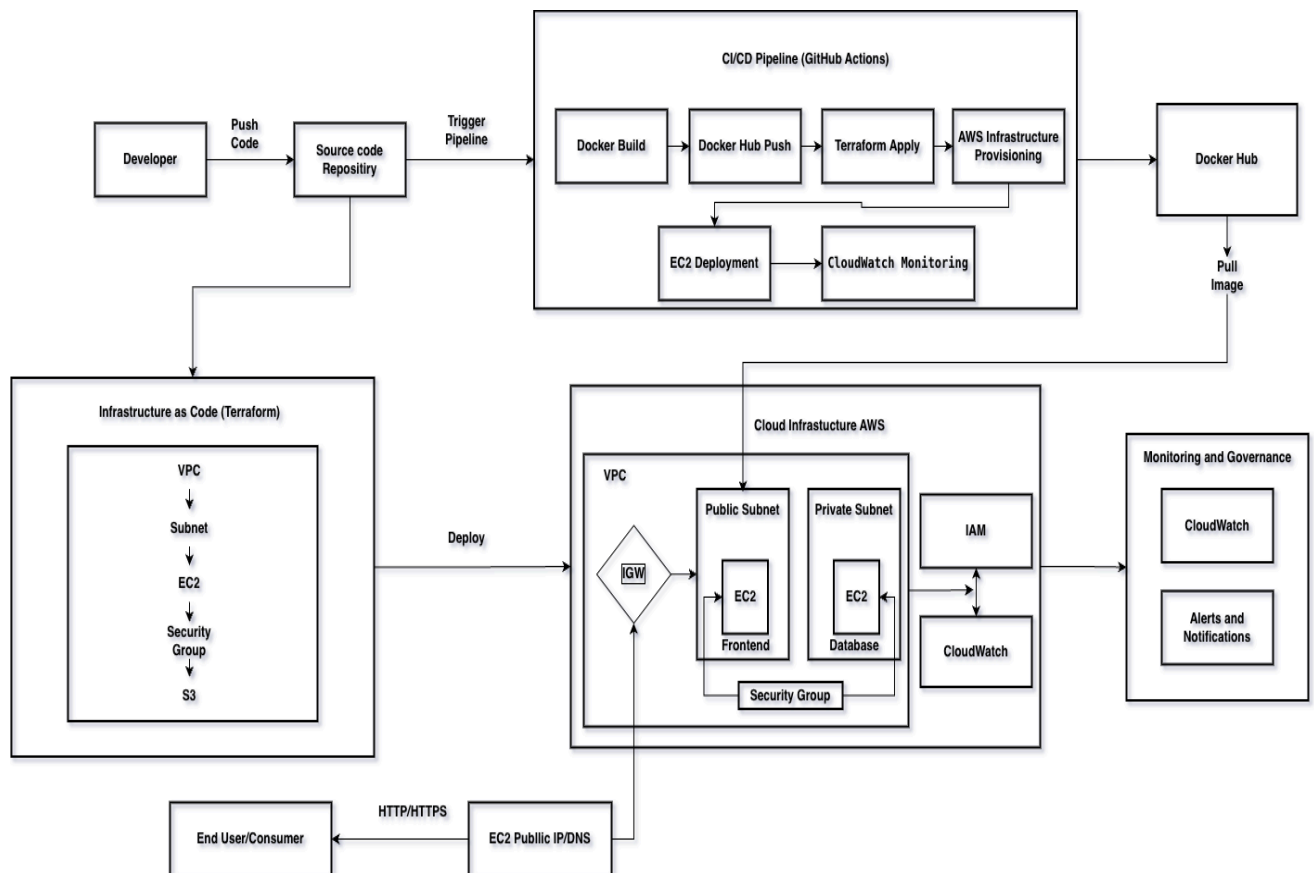
The system uses Docker, Terraform, AWS Cloud services, and GitHub Actions to automate the complete deployment workflow.

2. Platform Objectives

The primary objectives of the platform are:

- Automate application deployment
- Simplify infrastructure provisioning
- Standardize deployment workflows
- Provide reusable Infrastructure as Code templates
- Enable CI/CD automation
- Implement cloud monitoring and observability
- Reduce manual deployment effort

3. Platform Architecture Overview



4. Platform Capabilities Catalog

Platform Capability	Description
CI/CD Automation	Automated deployment pipeline using GitHub Actions
Infrastructure Provisioning	AWS infrastructure created using Terraform
Dockerized Deployment	Application deployment using Docker containers
Cloud Monitoring	AWS CloudWatch alarms and dashboard monitoring
Remote Terraform State	Centralized Terraform state management using AWS S3
Immutable Deployment	Infrastructure recreated consistently during deployment
Automated EC2 Deployment	EC2 automatically pulls and runs Docker container
Observability Hooks	CPU monitoring and EC2 status monitoring
Pipeline Automation	Automatic deployment after code push
Reusable Templates	Reusable Terraform and CI/CD configurations

5. Technology Stack

Technology	Purpose
Node.js + Express.js	Backend Application
Docker	Containerization
Docker Hub	Container Registry
Terraform	Infrastructure as Code
Amazon EC2	Application Hosting
Amazon VPC	Networking

Amazon CloudWatch	Monitoring and Observability
Amazon S3	Terraform Remote State
GitHub Actions	CI/CD Automation
GitHub	Source Code Management

6. Infrastructure Components

The following AWS infrastructure components are provisioned automatically using Terraform:

- Virtual Private Cloud (VPC)
- Public Subnet
- Internet Gateway
- Route Table
- Security Groups
- EC2 Instance
- CloudWatch CPU Alarm
- EC2 Status Check Alarm
- CloudWatch Dashboard
- S3 Remote Backend

7. Monitoring and Observability

Monitoring is implemented using Amazon CloudWatch.

Implemented monitoring features:

- CPU Utilization Alarm
- EC2 Status Check Alarm
- CloudWatch Dashboard
- Infrastructure health monitoring
- Network traffic monitoring

The CloudWatch dashboard provides visualization for:

- CPU Utilization
- Network In
- Network Out
- EC2 Status Checks

8. CI/CD Pipeline Workflow

The deployment workflow is fully automated.

Pipeline Steps:

- Developer pushes code to GitHub
- GitHub Actions workflow triggers automatically
- Docker image is built
- Docker image is pushed to Docker Hub
- Terraform provisions infrastructure
- Amazon EC2 pulls latest Docker image
- Docker container starts automatically
- Amazon CloudWatch monitoring becomes active

9. Security and Governance

The platform uses:

- GitHub Secrets for credential management
- Terraform Infrastructure as Code for controlled provisioning
- Docker container isolation
- Amazon VPC Security Groups for network access control
- Remote Terraform state management using encrypted Amazon S3 backend

10. Benefits of the Platform

- Faster deployment process
- Reduced manual infrastructure management
- Standardized deployment workflow
- Improved deployment consistency
- Infrastructure automation
- Improved monitoring and visibility
- Scalable DevOps foundation

11. Future Enhancements

Possible future improvements:

- API Gateway integration
- Kubernetes deployment
- Auto Scaling support
- HTTPS and Load Balancer
- IAM role-based access control
- Advanced monitoring dashboards